Benchmark : Open Source vs. Closed Source OCR Models

1 Introduction

This document presents a detailed performance comparison of Open Source and Closed Source OCR models.

2 Detailed Comparison by Dataset

2.1 1. Handwritten Dataset

Performance on handwritten text.

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	31.03	0.695	1.052	1.154
Tesseract	Open	38.44	0.634	0.987	**0.718**
TrOCR	Open	**70.75**	**0.295**	**0.346**	0.276
DocTR	Open	41.36	0.589	0.971	1.171
Google Vision	Closed	79.17	0.219	0.385	**0.301**
Gemini 2.5	Closed	**94.63**	**0.055**	**0.180**	2.568
GPT-4o	Closed	92.48	0.077	0.246	3.335
Claude 4	Closed	92.56	0.077	0.204	3.633

Table 1: Handwritten Dataset Results

2.2 2. Multilingual Datasets

2.2.1 2.1. English

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	89.82	0.102	0.332	2.990
Tesseract	Open	88.42	0.131	0.148	**0.895**
TrOCR	Open	0.90	0.991	0.999	0.270
DocTR	Open	**96.01**	**0.040**	**0.112**	2.900
Google Vision	Closed	**95.13**	**0.049**	**0.046**	**0.342**
Gemini 2.5	Closed	92.57	0.078	0.064	3.879
GPT-4o	Closed	91.13	0.091	0.087	8.708
Claude 4	Closed	93.05	0.071	0.075	14.496

Table 2: English Dataset Results

2.2.2 2.2. French

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	57.12	0.433	0.608	2.797
Tesseract	Open	55.54	0.452	0.568	**1.359**
TrOCR	Open	0.43	0.996	0.999	0.246
DocTR	Open	**58.93**	**0.415**	**0.555**	3.177
Google Vision	Closed	**56.90**	0.435	0.512	**0.349**
Gemini 2.5	Closed	57.18	0.436	0.510	3.567
GPT-4o	Closed	**57.78**	**0.464**	**0.529**	7.303
Claude 4	Closed	54.66	0.538	0.536	15.602

Table 3: French Dataset Results

2.2.3 2.3. Arabic

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	**38.58**	**0.638**	**0.931**	2.844
Tesseract	Open	21.38	0.793	0.929	**0.824**
TrOCR	Open	0.30	0.997	1.000	0.170
DocTR	Open	15.24	0.859	0.999	1.530
Google Vision	Closed	**52.09**	0.513	**0.720**	**0.304**
Gemini 2.5	Closed	**52.32**	**0.752**	0.714	4.288
GPT-4o	Closed	43.25	0.581	0.772	4.448
Claude 4	Closed	46.26	0.633	0.832	10.485

Table 4: Arabic Dataset Results

2.2.4 2.4. Japanese

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	**37.68**	**0.645**	1.056	3.912
Tesseract	Open	15.37	0.907	1.051	**1.077**
TrOCR	Open	1.60	0.984	0.997	0.261
DocTR	Open	26.39	0.998	1.783	1.295
Google Vision	Closed	50.19	0.574	0.758	**0.376**
Gemini 2.5	Closed	**55.82**	**0.466**	**0.609**	3.881
GPT-4o	Closed	49.04	0.529	0.627	3.190
Claude 4	Closed	50.66	0.501	0.638	4.215

Table 5: Japanese Dataset Results

2.2.5 2.5. German

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	52.02	0.493	0.787	3.093
Tesseract	Open	20.42	0.977	1.382	**1.232**
TrOCR	Open	4.04	0.960	0.996	0.296
DocTR	Open	**53.25**	**0.512**	**1.057**	1.671
Google Vision	Closed	**69.02**	0.326	**0.439**	**0.396**
Gemini 2.5	Closed	**69.14**	**0.350**	0.431	3.834
GPT-4o	Closed	59.86	0.408	0.486	3.767
Claude 4	Closed	62.72	0.379	0.450	6.517

Table 6: German Dataset Results

2.2.6 2.6. Spanish

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	**58.72**	**0.422**	**0.674**	0.341
Tesseract	Open	17.45	0.828	0.918	**0.314**
TrOCR	Open	6.98	0.930	0.998	0.143
DocTR	Open	53.60	0.513	0.904	1.080
Google Vision	Closed	65.07	0.367	0.409	**0.210**
Gemini 2.5	Closed	66.56	0.382	0.379	2.202
GPT-4o	Closed	67.96	0.344	**0.324**	2.221
Claude 4	Closed	**69.29**	**0.323**	0.337	3.550

Table 7: Spanish Dataset Results

2.2.7 2.7. Overlap Dataset

Performance on images with overlapping text layers.

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	22.50	1.160	1.444	3.732
Tesseract	Open	22.10	1.172	1.430	**1.304**
TrOCR	Open	0.30	0.997	1.000	0.224
DocTR	Open	**23.37**	**1.145**	**1.412**	2.829
Google Vision	Closed	22.13	1.185	1.440	**0.383**
Gemini 2.5	Closed	22.61	1.185	1.430	4.623
GPT-4o	Closed	22.25	1.175	**1.424**	8.412
Claude 4	Closed	**22.65**	**1.178**	1.428	13.460

Table 8: Overlap Dataset Results

2.3 2.8. Combined Dataset

Performance on the combined dataset (mixed languages, printed/handwritten, overlapping).

Model	Type	Accuracy (%)	CER ↓	WER ↓	Speed (s) ↓
EasyOCR	Open	24.08	0.759	1.649	9.50
Tesseract	Open	18.61	0.814	1.166	**1.68**
TrOCR	Open	-	-	-	-
DocTR	Open	**25.06**	**0.749**	**1.440**	2.54
Google Vision	Closed	34.77	0.652	1.831	**0.28**
Gemini 2.5	Closed	**37.45**	**0.626**	1.783	8.81
GPT-4o	Closed	20.62	0.794	**1.085**	5.92
Claude 4	Closed	25.89	0.741	1.709	14.48

Table 9: Combined Dataset Results

3 Average Speed Comparison

Summary of average processing speed across all datasets.

Model	Type	Average Speed (s) ↓
EasyOCR	Open	2.608
Tesseract	Open	**0.965**
TrOCR	Open	0.235
DocTR	Open	1.956
Google Vision	Closed	**0.332**
Gemini 2.5	Closed	3.605
GPT-4o	Closed	5.173
Claude 4	Closed	8.995

Table 10: Overall Average Speed

4 Final Summary: Best in Class

Criterion	Best Open Source Model	Best Closed Source Model
Handwritten	**TrOCR** (70.75%)	**Gemini 2.5** (94.63%)
English	**DocTR** (96.01%)	**Google Vision** (95.13%)
French	**DocTR** (58.93%)	**GPT-4o** (57.78%)
Arabic	**EasyOCR** (38.58%)	**Gemini 2.5** (52.32%)
Japanese	**EasyOCR** (37.68%)	**Gemini 2.5** (55.82%)
German	**DocTR** (53.25%)	**Gemini 2.5** (69.14%)
Spanish	**EasyOCR** (58.72%)	**Claude 4** (69.29%)
Overlap	**DocTR** (23.37%)	**Claude 4** (22.65%)
Combined	**DocTR** (25.06%)	**Gemini 2.5** (37.45%)
Average Speed	**Tesseract** (0.965s)	**Google Vision** (0.332s)

Table 11: Best by Category